

UPSC IES Electronics & Telecomm. Engineering Syllabus 2020

UPSC IES ECE Syllabus 2020

Meta Title - Read the Syllabus for UPSC IES ECE 2020 exam.

Meta Description - Get the detailed UPSC IES ECE Syllabus for 2020 with weightage. Get the detailed section-wise syllabus for UPSC IES ECE Prelims and Mains 2020.

FK - UPSC IES ECE Syllabus

Union Public Service Commission (UPSC) is conducting a national level exam in 2020 to recruit Electronics & Telecommunication Engineers. As per the official notification, there are 116 vacancies to be filled under various departments of the Indian Government. The selection of candidates is done in three stages - Prelims, Mains, and interviews. Candidates need to score the minimum cut off marks in both prelims and mains before being invited for the interview process. In the prelims, you have to solve two papers consisting of 250 objective type questions for a total of 500 marks. On the other hand, you have to solve 10 descriptive type questions in the mains for a total of 600 marks. The prelims results were declared on 20th February 2020 and qualified candidates are now preparing for the mains exam which is scheduled for 28th June 2020. Knowing the syllabus of each exam is important for every candidate, therefore, here is the complete UPSC IES ECE Syllabus 2020.

UPSC IES ECE 2020 Prelims Syllabus

The Electronics & Telecommunication Engineering (ECE) syllabus includes topics from two sections - Paper-I & Paper-II. Every aspiring candidate has to prepare General Studies & Engineering Aptitude topics for paper-I and ECE topics for paper-II. It is mandatory for you to qualify in prelims to become eligible for the mains exam. You need to score the minimum cut off marks out of 500 which will be declared by the UPSC. Furthermore, you will get the syllabus for both paper-I and paper-II of UPSC IES ECE in this article.

UPSC IES ECE 2020 Prelims Paper-I Syllabus

The first paper of the prelims exam is common for every branch which is General Studies & Engineering Aptitude. Candidates need to prepare every important topic for this exam as it holds a weightage of 200 marks. You will have to solve 100 objective type questions in this exam knowing there's a negative marking of one-third mark for every wrong answer. Here are the important topics for the UPSC IES ECE Prelims Paper-I syllabus.

- Current affairs - domestic & international
- Social conditions
- Economic theories and policies
- Industrial development

- Logical reasoning
- Analytical ability
- Standards and Quality practices
- Importance of Safety
- ICT based tools and applications
- Principles of Design
- Numerical Analysis
- Engineering Drawing
- Engineering Mathematics
- Ethics and values in the Engineering field
- Basics of Energy and Environment
- Basics of Project Management
- Basics of Material Science and Engineering

UPSC IES ECE 2020 Prelims Paper-II Syllabus

This paper consists of 150 objective type questions on Electronic & Telecommunication Engineering for a total of 300 marks. Various topics that you have studied in your ECE curriculum will be included in the syllabus for this exam. So, you need to study every topic that matters the most in ECE to secure good marks in this section. Here is the Prelims Paper-II syllabus for UPSC IES ECE 2020.

- Basic Electronics Engineering
- Basic Electrical Engineering
- Materials Science
- Electronic Measurements and Instrumentation
- Network Theory
- Analog and Digital Circuits
- Analog and Digital Communication Systems
- Control Systems
- Computer Organization and Architecture
- Electro Magnetism
- Advanced Electronics Topics
- Advanced Communication Topics

UPSC IES ECE 2020 Mains Syllabus

We have covered the overview for the prelims syllabus, let's see the overview of the mains exam. The total weightage of this exam is 600 marks. You need to solve 5 descriptive questions each from paper-I and paper-II in the mains exam. All the questions will be from the Electronics & Communication Engineering curriculum with no deduction for any wrong answers. Get the syllabus for the UPSC IES ECE Mains in the following sections.

UPSC IES ECE 2020 Mains Paper-I Syllabus

This is the first section in the UPSC IES ECE mains exam which will be conducted for 300 marks. There will be 8 descriptive questions in the paper out of which, the candidate has to solve only 5 questions. Each question will be based on various ECE topics that were studied by the candidate during his/her engineering course. Your preparations for this exam must include the following topics -

Syllabus for Basic Electronics Engineering:

- Basics of semiconductors
- Diode/Transistor basics and characteristics
- Diodes for different uses
- Junction & Field Effect Transistors (BJTs, JFETs, MOSFETs)
- Transistor amplifiers of different types, oscillators and other circuits
- Basics of Integrated Circuits (ICs)
- Bipolar, MOS and CMOS ICs
- Basics of linear ICs, operational amplifiers and their applications- linear/non-linear
- Optical sources/detectors
- Basics of Optoelectronics and its applications.

Syllabus for Basic Electrical Engineering:

- DC circuits ohm's & Kirchoff's laws, mesh and nodal analysis, circuit theorems
- Electromagnetism, Faraday's & Lenz's laws, induced EMF and its uses
- Single-phase AC circuits
- Transformers and its efficiency
- Basics - DC machines, induction machines, and synchronous machines
- Electrical power sources - basics: hydroelectric, thermal, nuclear, wind, solar
- Basics of batteries and their uses.

Syllabus for Materials Science:

- Electrical Engineering materials
- Crystal structure & defects
- Ceramic materials-structures, composites, processing and uses
- Insulating laminates for electronics, structures, properties and uses
- Magnetic materials, basics, classification, ferrites, Ferro/para-magnetic materials and components

- Nano materials-basics, preparation, purification, sintering, nanoparticles and uses
- Nano-optical/magnetic/electronic materials and uses
- Superconductivity uses.

Syllabus for Electronic Measurements and Instrumentation:

- Principles of measurement, accuracy, precision and standards
- Analog and Digital systems for measurement, measuring instruments for different applications
- Static/dynamic characteristics of measurement systems, errors, statistical analysis and curve fitting
- Measurement systems for non-electrical quantities
- Basics of telemetry
- Different types of transducers and displays
- Data acquisition system basics.

Syllabus for Network Theory:

- Network graphs & matrices
- Wye-Delta transformation
- Linear constant coefficient differential equations - time-domain analysis of RLC circuits
- 26 Solution of network equations using Laplace transforms - frequency domain analysis of RLC circuits
- 2-port network parameters - driving point & transfer functions
- State equations for networks
- Steady-state sinusoidal analysis.

Syllabus for Analog and Digital Circuits:

- Small signal equivalent circuits of diodes, BJTS and FETs
- Diode circuits for different uses
- Biasing & stability of BJT & JFET amplifier circuits
- Analysis/design of amplifier - single/multi-stage
- Feedback& uses Active filters, timers, multipliers, waveshaping, A/D-D/A converters
- Boolean Algebra & uses
- Logic gates, Digital IC families, Combinatorial/sequential circuits
- Basics of multiplexers, counters/registers/ memories /microprocessors, design & applications.

UPSC IES ECE 2020 Mains Paper-II Syllabus

Similar to the previous paper, you have to answer 5 out of the 8 descriptive questions available in paper-II. You have to prepare various topics from your ECE curriculum to get a good score out of 300 marks. So, here is the syllabus for Paper-II of the UPSC IES ECE mains exam 2020P

Syllabus For Analog and Digital Communication Systems:

- Random signals, noise, probability theory, information theory
- Analog versus digital communication & applications:
- Systems - AM, FM, transmitters/receivers, theory/practice/ standards, SNR comparison
- Digital communication basics: Sampling, quantizing, coding, PCM, DPCM, multiplexing-audio/video
- Digital modulation: ASK, FSK, PSK Multiple access: TDMA, FDMA, CDMA
- Optical communication: fibre optics, theory, practice/standards.

Syllabus For Control Systems:

- Classification of signals and systems
- Application of signal and system theory
- System realization
- Transforms & their applications
- Signal flow graphs, Routh-Hurwitz criteria, root loci, Nyquist/Bode plots
- Feedback systems - open & close loop types, stability analysis, steady-state, transient and frequency response analysis
- Design of control systems, compensators, elements of lead/lag compensation
- PID and industrial controllers

Syllabus For Computer Organization and Architecture:

- Basic architecture, CPU, I/O organisation, memory organisation, peripheral devices, trends
- Hardware /software issues
- Data representation & Programming
- Operating systems-basics, processes, characteristics, applications

- Memory management, virtual memory, file systems, protection & security Databases, different types, characteristics and design Transactions and concurrency control
- Elements of programming languages and its typical examples.

Syllabus for Electro Magnetics:

- Elements of vector calculus, Maxwell's equations - basic concepts
- Gauss', Stokes' theorems
- Wave propagation through different media
- Transmission Lines - different types, basics, Smith's chart, impedance matching/transformation, S-parameters, pulse excitation, use
- Waveguides-basics, rectangular types, modes, cut-off frequency, dispersion, dielectric types
- Antennas-radiation pattern, monopoles/dipoles, gain, arrays-active/passive, theory, use.

Syllabus for Advanced Electronics Topics:

- VLSI technology: Processing, lithography, interconnects, packaging, testing
- VLSI Design: Principles, MUX/ROM/PLA-based design, Moore & Mealy circuit design Pipeline concepts & functions
- Design for testability, examples
- DSP: Discrete-time signals/systems, uses
- Digital filters: FIR/IIR types, design, speech/audio/radar signal processing uses
- Microprocessors & microcontrollers, basics, interrupts, DMA, instruction sets, interfacing Controllers & uses Embedded systems.

Syllabus For Advanced Communication Topics:

- Communication networks: Principles /practices /technologies /uses /OSI model/security
- Basic packet multiplexed streams/scheduling
- Cellular networks, types, analysis, protocols (TCP/TCP/IP)
- Microwave & satellite communication: Terrestrial/space type LOS systems, block schematics link calculations, system design
- Communication satellites, orbits, characteristics, systems, use
- Fibre-optic communication systems, block schematics, link calculations, system design.

We have covered the syllabus for both prelims and mains exam of UPSC IES ECE 2020. Time to go through the exam pattern for this exam which is essential for every candidate to be aware of.

UPSC IES ECE 2020 Exam Pattern

Both prelims and mains exams will be conducted to filter candidates for the final interview process. There are two papers each in prelims as well as mains exam for 500 marks and 600 marks respectively. In the prelims exam, there will be 250 objective type questions and the mains exam will consist of 10 descriptive questions. We have provided exam patterns for both - prelims and mains in the following section.

UPSC IES ECE 2020 Prelims Exam Pattern

We have gone through the syllabus for the prelims exam but it is important to go through the exam pattern as well. This OMR-based exam consists of 2 sections - paper-I and paper-II in which there will be 250 objective questions to solve. Each question will carry 2 marks with a negative marking of one-third mark for every wrong attempt on the OMR sheet.

Sections	Questions	Marks	Duration
Paper-I: General Studies and Engineering Aptitude	100 MCQs	200	2 hours.
Paper-II: Electronics and Telecommunication Engineering	150 MCQs	300	3 hours

UPSC IES ECE 2020 Mains Exam Pattern

Only those candidates who have qualified in the prelims will appear for the UPSC IES ECE mains. In this exam, you will solve 5 descriptive questions each in both paper-I and paper-II. The total weightage of this exam is 600 marks with no negative marking system in place. Here is the exam pattern for UPSC ICES ECE 2020 mains.

Sections	Questions	Marks	Duration
Paper-I: Electronics & Telecommunication Engg.	Solve 5 out of 8 available questions	300	3 hours.
Paper-II: Electronics and Telecommunication Engg.	Solve 5 out of 8 available questions	300	3 hours

Download the UPSC IES ECE Syllabus (PDF Format) - [Click here](#)

We hope our article on UPSC ICE ECE Syllabus was informative and helpful. Do comment on your questions in the comment section below and we will surely answer your queries in no time. The mains exam is just around the corner and candidates should not afford to miss any opportunity to improve their preparations efforts. Get all the competitive exam's Syllabus PDFs, mock tests, online lectures, and lots more only on the Testbook [app](#). Install this app on your android device now and unlock many exciting offers - [Click here](#).

UPSC IES ECE Syllabus FAQs

Question.1 - What is the syllabus for UPSC IES ECE 2020?

Answer.1 - There are 2 written exams - Prelims and Mains which every candidate must qualify. You must prepare properly for paper-I and paper-II in both prelims and mains exam.

Question.2 - What is the UPSC IES ECE Syllabus for Prelims Paper-I?

Answer.2 - The candidate has to study various topics such as Current affairs, Social conditions. industrial development, economic policies, etc.

Question.3 - What is the UPSC IES Prelims ECE Syllabus for Paper-II?

Answer.3 - There will be various Electronics & Telecommunication engineering topics such as Materials Science, Electronic Measurements and Instrumentation, Network Theory, Analog, and Digital Circuits, Analog and Digital Communication Systems, etc.

Question.4 - What is the UPSC IES Mains ECE Syllabus for Paper-I?



Answer.4 - For the mains paper-I of this exam, you need to study topics such as Basic Electronics Engineering, Basic Electrical Engineering, Materials Science, Electronic Measurements and Instrumentation, Network Theory, etc.

Question.5 - What is the UPSC IES Mains ECE Syllabus for Paper-II?

Answer.5 - Candidates have to study topics such as Control Systems, Computer Organization, and Architecture, Electro Magnetics, Advanced Electronics, etc.

